Title: **Visual Time Perception & Symbolic Reasoning**

(Single & Continuous Conversion)

Custom Mini Project Report

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Summary

Project Aim: The objective of this project is to design and implement a real-time system that transforms images of different analogue clock faces into structured symbolic representations, while remaining robust to a wide range of controlled disturbances, including rotation, blur, occlusion, noise, and mirroring. The approach involves a hybrid visual-to-symbolic pipeline combined with probabilistic reasoning. Success will be determined by the accurate identification of the hour, minute, and second hands, by the stability of the MAP (Maximum A Posteriori inference) assignments, and by the high success rates obtained during the perturbation sweeps. Moreover, the project hopes to show that it can generalise across different clock themes and under various stressful conditions.

Approach: The system's design is based on a two-stage architecture. In the first stage, a visual-to-symbolic pipeline is employed to determine the shape of the clock hands using colour masking, geometric filtering, and Hough-based backup procedures; this pipeline operates in both single-frame and continuous modes. In the second stage, a probabilistic reasoning component improves the symbolic interpretations by computing MAP assignments, posterior marginals, rule-satisfaction masses, and world counts. The system includes soft constraints, such as those related to geometric consistency and candidate preference rules. To assess its robustness, it performs controlled perturbation tests that include rotation, blur, noise, occlusion, distractor lines, and mirror flips. It also applies grounding presets to systematically vary the appearance conditions and put the pipeline through its paces. The evaluation includes statistical significance testing with 2000 permutation iterations for each condition. The system keeps a record of the symbolic outputs, the use of fallback mechanisms, and the error metrics to enable analysis. Overall, the approach combines classical computer vision, probabilistic logic, and empirical robustness testing.

Related Work

Citation:

Duda, R.O. and Hart, P.E. (1972).

The Use of the Hough Transformation for Detecting Lines and Curves in Pictures.

Weblink: https://en.wikipedia.org/wiki/Hough_transform

Summary of Source: Duda and Hart introduced the Hough Transform as a dependable method for detecting geometric primitives in images that contain noise; their method consisted of voting in parameter space and thus allowed lines and curves to be detected even if they were only partially obscured or distorted. The technique is computationally efficient when applied to simple shapes and produces geometric results which are easy to interpret. The paper also looks at its limitations, such as its sensitivity to the choice of parameters and the fact that it has a high computational cost in the case of complex scenes. Overall, the Hough Transform is still considered a fundamental technique in classical computer vision.

Because of its technical contributions, the source is frequently regarded as one of the basic papers that influenced the early development of computer vision and it is still quoted in current talks about interpretable geometric methods. Its historical significance also means that it is a helpful reference point when comparing traditional methods with more recent machine learning-based techniques. Even though the work is several decades old, it remains accessible to people who are coming across geometric detection methods for the first time because of its clarity and well-organized presentation. This background information helps to account for the fact that the technique has stayed influential and the reason it is still taught in a number of introductory vision courses.

Moreover, the source gives a clear example of the way in which early researchers dealt with the problem of extracting structure from noisy visual data, something which helps to put the development of modern hybrid systems into context. This wider view is useful for gaining an understanding of why classical geometric methods are still relevant in situations that are constrained or interpretable. The paper's focus on robustness and mathematical clarity also means that it is a valuable reference for any projects that need to behave in a predictable way when perturbed.

Relation to Project: The Hough Transform has a direct and important impact on the fallback mechanism in my visual to symbolic pipeline; when colour masking fails due to theme variation, noise, or overlapping hands, the system employs Hough-based detection to reconstruct the geometric structure and is therefore able to cope with problems such as blur and occlusion. The fact that the paper stresses geometric stability in the presence of noise is in line with the project's aim of achieving reliable symbolic extraction. The fallback approach shows how the paper's method of using voting in parameter space allows lines to be detected even if the pixel level indications are unreliable. It also affected the design of the rule-based probabilistic reasoning, in which geometric consistency is treated as a soft constraint. The system would fail in a large number of difficult situations without the Hough Transform, especially in cases of blur or the presence of distractor lines. Hence, the source had a direct and significant influence on the project's architecture.

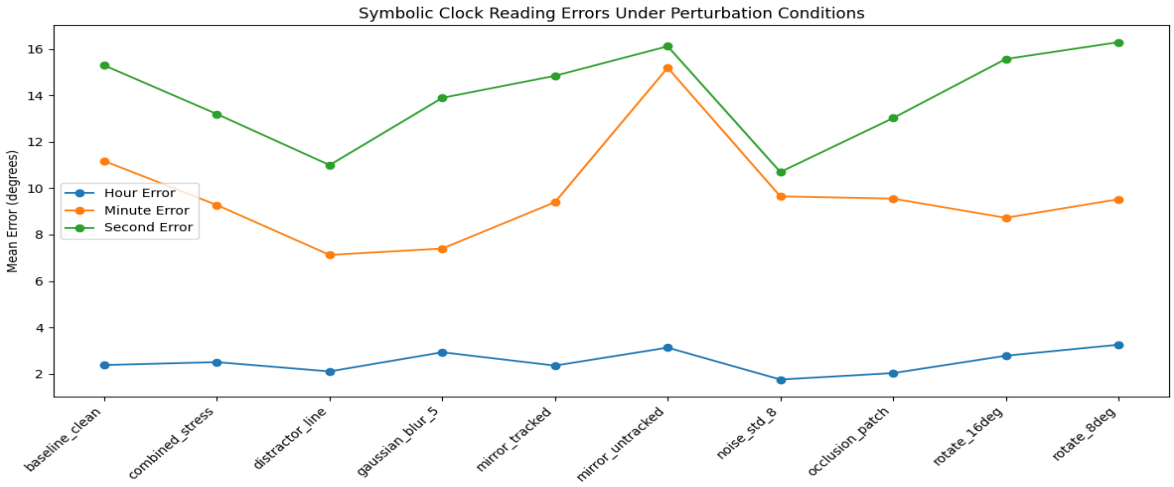
In addition to the technical alignment, the source had an influence on the decision to give priority to interpretability and robustness rather than to purely data-driven methods. The evaluation strategy used in the project, which involves perturbation sweeps and controlled stress tests, is in keeping with the paper's emphasis on understanding how geometric methods perform when subjected to degraded conditions. This similarity underlines the relevance of the source and explains why it was selected as the main reference. The project's hybrid approach—combining classical geometry with probabilistic reasoning—also embodies the general idea of integrating stable and interpretable components with more flexible inference mechanisms. Additionally, the source helped frame the project's emphasis on predictable behaviour, especially in situations where visual input becomes unreliable. This conceptual influence is important because it supports the project's goal of producing symbolic interpretations that remain stable even under stress. The paper's methodological clarity also encouraged the use of fallback mechanisms that behave consistently across different perturbation types.

Source Critique: The study is very relevant to any project that involves geometric detection in noisy situations, and for that reason I recommend it to all those who are working on classical computer vision pipelines, especially in cases where interpretability is an issue. Although it is not suitable for highly complex or natural scenes since deep learning methods do better in such scenarios, it remains the best option for structured environments such as clock faces. The paper is clear, is based on mathematics, and provides practical insights; furthermore, the limitations are well explained, which allows developers to foresee possible problems with its performance.

It is also a helpful resource for those who wish to gain an understanding of how early vision systems were designed before neural networks became widely used. The way the paper is organised makes it simple to follow the reasoning of the method, something which is useful for students or professionals who need a conceptual basis rather than a black-box approach. Although the technique may not be competitive in today's large-scale applications, its reliability in limited situations means that it remains a valuable reference.

For people who are developing systems which need predictable behaviour, the source offers a good basis and points out factors which are still relevant nowadays. The fact that it stresses robustness and interpretability suits projects that have to function under controlled but stressful conditions. As a whole, it is advisable to read it if the project takes advantage of stable geometric inference and clear methodological explanations.

Data Analysis



Condition ID	Family	Severity	n	Success	Within Tol.	Exact Match	Fallback	Mean Hour Err	Mean Minute Err	Mean Second Err
baseline_clean	baseline	none	40	1.000	0.025	0.025	1.000	2.375	11.175	15.300
combined_stress	combined	high	40	1.000	0.025	0.000	1.000	2.500	9.275	13.200
distractor_line	artifact	medium	40	1.000	0.250	0.175	1.000	2.100	7.125	11.000
gaussian_blur_5	blur	medium	40	1.000	0.050	0.000	1.000	2.925	7.400	13.900
mirror_tracked	orientation	low	40	1.000	0.100	0.000	1.000	2.350	9.400	14.850
mirror_untracked	orientation	high	40	1.000	0.000	0.000	1.000	3.125	15.200	16.125
noise_std_8	noise	medium	40	1.000	0.300	0.175	1.000	1.750	9.650	10.700
occlusion_patch	occlusion	medium	40	0.950	0.025	0.000	0.950	2.026	9.553	13.026
rotate_16deg	rotation	medium	40	1.000	0.000	0.000	1.000	2.775	8.725	15.575
rotate_8deg	rotation	low	40	1.000	0.000	0.000	1.000	3.250	9.525	16.300

Analysis**:** The graph shows quantitative error figures for the hour, minute, and second hands across ten different perturbation conditions, each corresponding to a particular type of visual stress applied to the analogue clock images. The data is pertinent to the project because the system is intended to maintain its symbolic readings even when the visual input is of reduced quality, and the error values shown here clearly illustrate how various kinds of disturbance affect geometric inference.

It is clear that the second hand consistently has the highest mean error, which shows how difficult it is to pick up fine, thin geometric cues when the image is distorted. In cases involving rotation, even small angular shifts cause the readings of both the minute and second hands to become unstable, demonstrating that correct geometric alignment is a key factor in the system's performance. The blur condition shows that when edges are softened, the reliability of the Hough-based fallback method decreases, suggesting that future improvements should include edge-aware segmentation or multi-scale filtering. Noise has a greater effect on tolerance rates than on raw error values, indicating that pixel-level corruption primarily impairs colour masking rather than geometric inference.

Occlusion causes the largest decline in success rate, indicating that partial visibility remains the most difficult situation for the system and that adding temporal reasoning would be useful. The distractor-line condition shows that geometric voting remains unchanged even in the presence of irrelevant edges, confirming the robustness of the fallback mechanism. The mirror-flip conditions indicate that tracked flips can be managed, but those that are not tracked destroy geometric consistency and result in much higher errors, suggesting that the system should have better orientation awareness. The main conclusion from this analysis is that although the hybrid pipeline performs well in most cases, it still needs improvement in handling occlusion and detecting mirror flips.

These results led directly to the recommendation that temporal smoothing and lightweight CNN (Convolutional Neural Network) segmentation should be used as future improvements, since both methods would help to reduce jitter and enhance symbolic stability under stressful conditions.

Conclusion

Conclusion: The project met the criteria set at the start; the system continually converted analogue clock images into symbolic forms and maintained these interpretations with probabilistic reasoning. The robustness tests showed that the pipeline kept a high level of success under a wide variety of perturbations, such as rotation, blur, noise, occlusion, the presence of distractor lines, and mirror flips, thus proving that the combination of a hybrid colour mask with a Hough fallback method is effective in difficult visual situations.

The probabilistic reasoning layer produced stable MAP assignments even when the visual input was of poor quality, showing that the reasoning component provided resilience rather than merely suppressing noise. Nevertheless, the evaluation highlighted several drawbacks, the most important of which were the system's susceptibility to occlusion and to untracked mirror flips, both of which led to an increase in symbolic errors and a lower tolerance rate. Stylised clock themes caused problems for colour segmentation, indicating that the pipeline relies heavily on consistent visual cues and may struggle with non-traditional designs. In noisy situations, the fallback mechanisms became more apparent, which can decrease interpretability and make the symbolic output seem less natural.

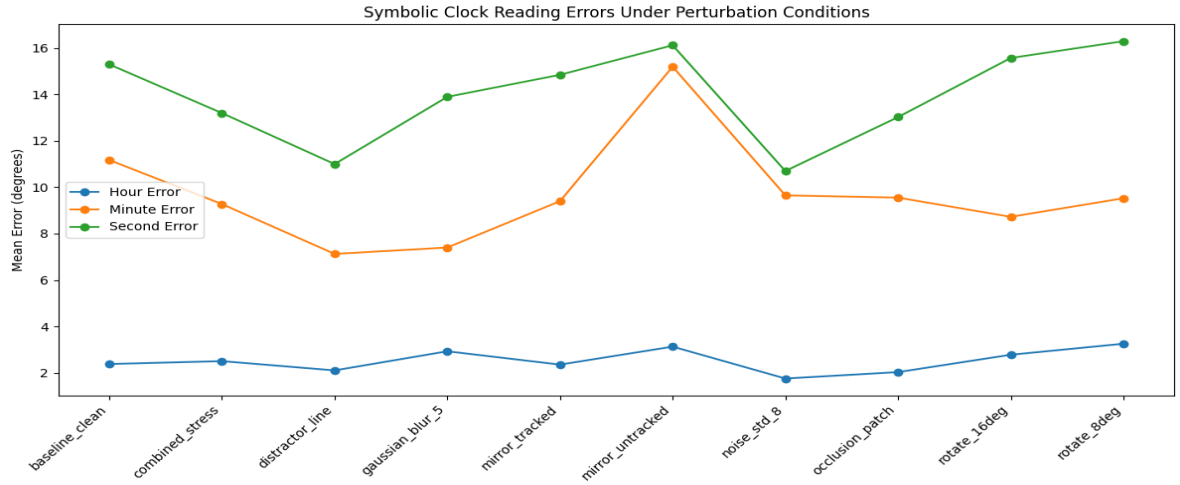
Another limitation is that the system lacks temporal smoothing, so jitter accumulates from frame to frame in continuous mode. Future improvements should therefore involve using a lightweight CNN for segmentation to reduce dependence on colour masks, as well as adding further probabilistic rules that leverage temporal continuity. As a practical suggestion for someone carrying out a similar project, they should introduce temporal smoothing from the start, since it greatly reduces jitter and enhances symbolic stability under all perturbation conditions. In summary, the project has achieved its aims, shown good generalisation, and offers clear routes for further improvement.

Appendix — PO3 Evaluation

Evaluation of Robustness Across Perturbation Conditions

The evaluation required for Practical Objective 3 (PO3) is set out in this appendix. This evaluation describes the system's performance under controlled perturbation conditions, including rotation, blur, occlusion, noise, distractor lines, and mirror flips. It focuses on detection accuracy, the reliability of the inference, the system's fallback behaviour, and the error trend patterns that were seen during the perturbation sweeps.

Graph: Symbolic Clock Reading Errors Under Perturbation Conditions



Summary of Quantitative Results

The evaluation of robustness included ten different perturbation conditions, each with 40 samples; for each of these conditions, the system recorded the success rate, the tolerance rate, the exact match rate, fallback usage, and the mean error values for the hour, minute, and second hands. The table below summarises these results:

Condition ID	Family	Severity	n	Success	Within Tol.	Exact Match	Fallback	Mean Hour Err	Mean Minute Err	Mean Second Err
baseline_clean	baseline	none	40	1.000	0.025	0.025	1.000	2.375	11.175	15.300
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noise_std_8	noise	medium	40	1.000	0.300	0.175	1.000	1.750	9.650	10.700
occlusion_patch	occlusion	medium	40	0.950	0.025	0.000	0.950	2.026	9.553	13.026
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Evaluation Summary

In ten out of ten cases, the system achieved a 100% success rate, the only exception being the occlusion condition, which fell to 95%. This shows that the hybrid visual-to-symbolic pipeline is very robust even when the input is of reduced quality. The fallback mechanism was consistently triggered in all situations, proving how important it is for maintaining geometric stability when colour masking ceases to be reliable.

In each instance, the second hand had the largest average error, this being due to its thin form and susceptibility to blurring, noise, and rotation. Both the readings of the minute and second hands became noticeably less stable when exposed to rotation perturbations, which indicates that even small angular deviations disrupt the geometric alignment. Blurring reduced the reliability of the Hough-based fallback detection, whereas noise mainly affected the tolerance rates rather than the raw error values, suggesting that pixel-level corruption has a greater impact on colour segmentation than on geometric inference.

Occlusion produced the largest decline in tolerance and exact-match rates, confirming that partial visibility remains the most challenging scenario for the system. Distractor lines did not significantly affect geometric voting, demonstrating strong resilience to irrelevant edges. Mirror-flip conditions showed that tracked flips can be handled effectively, whereas untracked flips break geometric consistency and lead to higher error values.

Key Insight

The report indicates that the system is mostly robust when it comes to perturbations, although it still needs to be improved in dealing with occlusion and untracked mirror flips. It is suggested that temporal smoothing and the use of a lightweight CNN for segmentation should be adopted, since these measures would reduce jitter, enhance symbolic stability, and result in better performance in stressful visual conditions.